

Cinema Management System

Analysis Report — Assignment 1

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1. Introduction

The Cinema Management System (CMS) is an application that automates the operations of a cinema, from maintaining the movie catalogue and scheduling showtimes to selling tickets, handling payments and producing management reports. It is used by the cinema's staff; customers do not operate the system.

A Manager maintains the reference data (movies, genres, cinema halls, showtimes, staff, snacks and promotions), publishes the showtimes (Post/Unpost), manages user roles and produces reports. A Cashier registers customers and sells tickets at the box office: creating a booking, selecting seats, applying promotions, taking payment, generating the ticket and validating tickets at the entrance. Because the manager is a senior member of staff, the Manager can also perform every Cashier task (generalisation Manager ▷ Cashier). The customer is a data entity (the customer table), not an actor.

The core operation is selling a ticket: the system shows the seats available for a chosen showtime, the seat is reserved, payment is taken and the ticket is issued; the seat is then marked sold so it can never be sold twice.

Actors: Manager (primary), Cashier (primary). **Generalisation:** Manager ▷ Cashier.

2. List of Features (Use Cases)

No	Use Case	Actor
1	Add Movie	Manager
2	Update Movie	Manager
3	Delete Movie	Manager
4	Search Movie by Title, Genre, Language	Cashier
5	Add Genre	Manager
6	Update Genre	Manager
7	Delete Genre	Manager
8	Search Genre by Name	Manager
9	Add Cinema Hall	Manager
10	Update Cinema Hall	Manager
11	Delete Cinema Hall	Manager
12	Search Cinema Hall by Name, Type, Capacity	Manager
13	Add Showtime	Manager
14	Update Showtime	Manager
15	Delete Showtime	Manager
16	Search Showtime by Date, Movie, Hall, Time	Cashier
17	Add Customer	Cashier
18	Update Customer	Cashier
19	Delete Customer	Cashier
20	Search Customer by Id, Name, Phone	Cashier
21	Add Staff	Manager
22	Update Staff	Manager
23	Delete Staff	Manager
24	Search Staff by Id, Name, Role	Manager
25	Add Booking	Cashier
26	Update Booking	Cashier
27	Delete Booking	Cashier
28	Search Booking by Id, Customer, Date, Showtime	Cashier
29	Add Ticket	Cashier
30	Update Ticket	Cashier

31	Delete Ticket	Cashier
32	Search Ticket by Code, Showtime, Customer	Cashier
33	Add Payment	Cashier
34	Update Payment	Cashier
35	Delete Payment	Cashier
36	Search Payment by Id, Date, Method, Customer	Cashier
37	Add Snack	Manager
38	Update Snack	Manager
39	Delete Snack	Manager
40	Search Snack by Name, Category	Cashier
41	Sell Snack	Cashier
42	Add Promotion	Manager
43	Update Promotion	Manager
44	Delete Promotion	Manager
45	Search Promotion by Code, Period	Cashier
46	Post Showtime	Manager
47	Unpost Showtime	Manager
48	Select Seat	Cashier
49	Apply Promotion to Booking	Cashier
50	Generate Ticket	Cashier
51	Validate Ticket at Entrance	Cashier
52	Issue Refund	Cashier
53	Generate Sales Report	Manager
54	Generate Hall Occupancy Report	Manager
55	Manage User Roles	Manager
56	Login	Cashier
57	Logout	Cashier

3. Use Case Diagrams

Replace each placeholder below with the matching PNG you exported from StarUML (File → Export Diagram As → PNG).

1 - Movie & Genre Management

[Insert StarUML diagram «1 - Movie & Genre Management» here]

2 - Hall & Showtime Management

[Insert StarUML diagram «2 - Hall & Showtime Management» here]

3 - Customer & Staff Management

[Insert StarUML diagram «3 - Customer & Staff Management» here]

4 - Booking, Ticket & Payment

[Insert StarUML diagram «4 - Booking, Ticket & Payment» here]

5 - Snack & Promotion

[Insert StarUML diagram «5 - Snack & Promotion» here]

6 - Reporting & Security

[Insert StarUML diagram «6 - Reporting & Security» here]

4. Use Case Specifications

Full specifications are given for the four CRUD templates (Add / Update / Delete / Search, illustrated on Movie) and for every business operation. All the other CRUD use cases — Update / Delete / Search of Genre, Cinema Hall, Showtime, Customer, Staff, Booking, Ticket, Payment, Snack and Promotion — follow exactly the same four templates and are not repeated here.

UC-01 — Add Movie	
Actor(s)	Manager
Description	Adds a new movie to the system.
Preconditions	The Manager is logged in.
Postconditions	A new movie record is stored.
Main Flow	1. Open the Movie form. 2. Enter the movie details. 3. Click Save. 4. The system validates the data and stores the new movie.
Alternative Flows	3a. A required field is missing or a duplicate exists → the system shows an error and nothing is saved.

UC-02 — Update Movie	
Actor(s)	Manager
Description	Modifies the details of an existing movie.
Preconditions	The Manager is logged in.
Postconditions	The movie record is updated.
Main Flow	1. Search and select the movie. 2. Edit the details. 3. Click Save. 4. The system validates and updates the record.
Alternative Flows	3a. Invalid data → error, no change is saved.

UC-03 — Delete Movie	
Actor(s)	Manager
Description	Removes a movie from the system.
Preconditions	The Manager is logged in.
Postconditions	The movie record is deleted.
Main Flow	1. Select the movie. 2. Confirm the deletion. 3. The system removes the record.
Alternative Flows	2a. The movie is referenced elsewhere → the deletion is blocked with a warning.

UC-04 — Search Movie by Title, Genre, Language	
Actor(s)	Cashier

Description	Finds movie records by title, genre, language.
Preconditions	The Cashier is logged in.
Postconditions	The matching records are displayed.
Main Flow	1. Enter the search criteria (Title, Genre, Language). 2. The system displays the matching movie records.
Alternative Flows	2a. No match → the system shows an empty result.

UC-25 — Add Booking	
Actor(s)	Cashier
Description	The cashier creates a booking for a showtime on behalf of a customer: selects seats, applies any promotion, takes payment and issues the ticket.
Preconditions	The cashier is logged in; a posted showtime with a free seat exists.
Postconditions	A booking, ticket and payment are saved; the seats are marked sold.
Main Flow	1. Select a posted showtime. 2. Select one or more seats («include» Select Seat). 3. Enter/select the customer. 4. (optional) Apply a promotion («extend» Apply Promotion). 5. Take payment («include» Add Payment). 6. Save the booking and issue the ticket («include» Generate Ticket); mark the seats sold. 7. Print/show the ticket.
Alternative Flows	2a. Seat already sold → choose another seat. 5a. Payment fails → release the seats, no ticket.

UC-48 — Select Seat	
Actor(s)	Cashier
Description	Chooses one or more available seats for a showtime.
Preconditions	A showtime is selected; a seat map is displayed.
Postconditions	The chosen seats are reserved for the booking.
Main Flow	1. The system shows available seats. 2. The cashier selects the seat(s). 3. The seats are held.
Alternative Flows	2a. Seat just taken → ask for another seat.

UC-33 — Add Payment	
Actor(s)	Cashier
Description	Charges the customer for a booking.
Preconditions	A booking with a total amount is awaiting payment.
Postconditions	A payment record is created (Paid or Failed).
Main Flow	1. Show the amount due and methods (Card, Mobile, Cash). 2. Choose a method. 3. Process payment, record as Paid.

Alternative Flows	3a. Declined → record Failed; booking stays unpaid.
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UC-50 — Generate Ticket	
Actor(s)	Cashier
Description	Issues the ticket(s) for a paid booking with a unique code.
Preconditions	The booking is paid.
Postconditions	Ticket(s) are created with status Valid and printed/shown.
Main Flow	1. Create a ticket per seat with a unique code. 2. Print/show the ticket(s).
Alternative Flows	—

UC-49 — Apply Promotion to Booking	
Actor(s)	Cashier
Description	Applies a valid promotion code to a booking to reduce the total.
Preconditions	A booking is in progress; a valid promotion exists.
Postconditions	The discount is applied and the total recalculated.
Main Flow	1. Enter the promotion code. 2. The system checks it is valid and in date. 3. Apply the discount.
Alternative Flows	2a. Invalid/expired code → no discount; show a message.

UC-52 — Issue Refund	
Actor(s)	Cashier
Description	Refunds a cancelled booking.
Preconditions	The booking is paid and being cancelled.
Postconditions	A refund payment is recorded; the ticket status becomes Refunded.
Main Flow	1. Select the cancelled booking. 2. Confirm the refund. 3. Record the refund and mark the ticket Refunded.
Alternative Flows	2a. Booking not paid → nothing to refund.

UC-51 — Validate Ticket at Entrance	
Actor(s)	Cashier
Description	Checks that a presented ticket is valid for the current showtime and marks it used.
Preconditions	The cashier is logged in; the ticket exists.
Postconditions	The ticket is marked Used; entry granted or refused.
Main Flow	1. Scan/enter the ticket code. 2. Find the ticket and showtime. 3. Check it is Valid, for the right showtime and not used. 4. Mark Used and show 'Access granted' with the seat.

Alternative Flows	3a. Already used / wrong showtime → 'Invalid ticket', refuse entry.
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UC-41 — Sell Snack	
Actor(s)	Cashier
Description	Sells one or more snacks to a customer and records the snack order.
Preconditions	The cashier is logged in; snacks exist in the menu.
Postconditions	A snack order is saved with its line(s) and total.
Main Flow	1. Select the snack(s) and quantity. 2. The system computes the total. 3. Take payment. 4. Save the snack order.
Alternative Flows	2a. Snack out of stock → choose another.

UC-46 — Post Showtime	
Actor(s)	Manager
Description	Publishes a showtime so it becomes available for selling tickets.
Preconditions	The manager is logged in; the showtime exists.
Postconditions	The showtime status becomes Posted.
Main Flow	1. Select the showtime. 2. Click Post. 3. The status becomes Posted.
Alternative Flows	—

UC-47 — Unpost Showtime	
Actor(s)	Manager
Description	Hides a showtime so tickets can no longer be sold for it.
Preconditions	The manager is logged in; the showtime is Posted.
Postconditions	The showtime status becomes Unposted.
Main Flow	1. Select the showtime. 2. Click Unpost. 3. The status becomes Unposted.
Alternative Flows	2a. Tickets already sold → warn before unposting.

UC-53 — Generate Sales Report	
Actor(s)	Manager
Description	Produces a report of tickets sold and revenue for a chosen period.
Preconditions	The manager is logged in.
Postconditions	A report is displayed/printed.
Main Flow	1. Choose the period. 2. The system totals tickets sold and revenue. 3. Display the report with a grand total.
Alternative Flows	1a. No data → zero totals.

UC-54 — Generate Hall Occupancy Report	
Actor(s)	Manager
Description	Shows, per showtime, the seats sold versus capacity and the occupancy rate.
Preconditions	The manager is logged in.
Postconditions	An occupancy report is displayed/printed.
Main Flow	1. Choose the period/hall. 2. The system computes seats sold / capacity per showtime. 3. Display the report.
Alternative Flows	—

UC-55 — Manage User Roles	
Actor(s)	Manager
Description	Assigns or changes the role (Manager/Cashier) of a staff account.
Preconditions	The manager is logged in.
Postconditions	The staff member's role is updated.
Main Flow	1. Select the staff account. 2. Choose the role. 3. Save.
Alternative Flows	—

UC-56 — Login	
Actor(s)	Cashier
Description	Authenticates a staff member and opens the menu according to their role.
Preconditions	The application is running; a valid staff account exists.
Postconditions	The staff member is authenticated with role-based access.
Main Flow	1. Enter username and password. 2. The system validates and reads the role. 3. Open the menu (full for Manager, selling for Cashier).
Alternative Flows	2a. Invalid credentials → show 'Invalid login credentials', stay on the login screen.

UC-57 — Logout	
Actor(s)	Cashier
Description	Ends the staff member's session securely.
Preconditions	A staff member is logged in.
Postconditions	The session is closed; the login screen is shown.
Main Flow	1. Click Logout. 2. The system closes the session and returns to the login screen.
Alternative Flows	—

5. Input Design (Forms / Dialog Boxes)

Hand sketches are acceptable. The fields of each form are listed below.

Form	Fields
Login	Username Password
Movie	Title Genre (dropdown) Duration Language Rating
Genre	Name
Cinema Hall	Hall Name Type (2D/3D) Capacity
Showtime	Movie (dropdown) Hall (dropdown) Date Time Ticket Price Status (Posted)
Customer	Full Name Phone Email
Staff	Full Name Username Password Role (Manager/Cashier)
Booking / Seat selection	Showtime (dropdown) Seat map Customer Promotion code Total
Payment	Amount Due Method (Card/Mobile/Cash)
Snack	Name Category Price (Manager manages the menu)
Sell Snack	Booking (dropdown) Snack (dropdown) Quantity Line Total
Promotion	Code Description Discount % Start Date End Date

6. Output Design (Screen / Paper)

Output	Content
Listing tables	one per entity: Movies, Genres, Halls, Showtimes, Customers, Staff, Bookings, Tickets, Payments, Snacks, Promotions
Ticket (printout)	cinema name, movie, date/time, hall, seat(s), price, ticket code/QR
Payment receipt	receipt no., amount, method, date
Sales Report	tickets sold and revenue for a period, grand total
Hall Occupancy Report	per showtime: seats sold / capacity / occupancy %
Error dialogs	invalid login, duplicate title, showtime clash, seat already sold

7. Database Design

staff

Attribute	Data Type	Key
staff_id	INT	PK
full_name	VARCHAR(100)	
username	VARCHAR(50)	
password	VARCHAR(255)	
role	VARCHAR(10)	Manager / Cashier

customer

Attribute	Data Type	Key
customer_id	INT	PK
full_name	VARCHAR(100)	
phone	VARCHAR(20)	
email	VARCHAR(100)	

genre

Attribute	Data Type	Key
genre_id	INT	PK
name	VARCHAR(50)	

movie

Attribute	Data Type	Key
movie_id	INT	PK
title	VARCHAR(120)	
genre_id	INT	FK -> genre
duration	INT	
language	VARCHAR(40)	
rating	VARCHAR(10)	

cinema_hall

Attribute	Data Type	Key
hall_id	INT	PK
hall_name	VARCHAR(60)	
hall_type	VARCHAR(20)	
capacity	INT	

showtime

Attribute	Data Type	Key
show_id	INT	PK
movie_id	INT	FK -> movie
hall_id	INT	FK -> cinema_hall
show_date	DATE	
show_time	TIME	
ticket_price	DECIMAL(8,2)	
status	VARCHAR(10)	Posted / Unposted

booking

Attribute	Data Type	Key
booking_id	INT	PK
customer_id	INT	FK -> customer
show_id	INT	FK -> showtime
promotion_id	INT	FK -> promotion (null)
booking_datetime	DATETIME	
status	VARCHAR(10)	Confirmed / Cancelled
total_amount	DECIMAL(8,2)	

ticket

Attribute	Data Type	Key
ticket_id	INT	PK
booking_id	INT	FK -> booking
seat_number	VARCHAR(8)	
price	DECIMAL(8,2)	
status	VARCHAR(10)	Valid / Used / Refunded

payment

Attribute	Data Type	Key
payment_id	INT	PK
booking_id	INT	FK -> booking
amount	DECIMAL(8,2)	
method	VARCHAR(10)	Card / Mobile / Cash
payment_datetime	DATETIME	
status	VARCHAR(10)	Paid / Refunded

snack

Attribute	Data Type	Key
snack_id	INT	PK
name	VARCHAR(50)	
category	VARCHAR(30)	
price	DECIMAL(8,2)	

snack_order

Attribute	Data Type	Key
order_id	INT	PK
booking_id	INT	FK -> booking
snack_id	INT	FK -> snack
quantity	INT	
line_total	DECIMAL(8,2)	

promotion

Attribute	Data Type	Key
promotion_id	INT	PK
code	VARCHAR(20)	
description	VARCHAR(100)	
discount_percent	DECIMAL(5,2)	
start_date	DATE	
end_date	DATE	